

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 2.1: Similarity and Enlargement — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.1: Similarity and Enlargement
Driving Question	DRIVING QUESTION: When you scale up a shape or solid — like doubling the size of a water tank or enlarging a building plan — what exactly happens to its lengths, its area, and its volume, and how can we use this to solve real problems in our community?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Anchoring Phenomenon & Properties of Similar Figures	Learners observed a school architectural plan labelled Scale 1:200 and two similar water tanks where one holds 8 times as much water despite appearing only twice as tall. Learners also measured corresponding sides and angles of everyday objects (Kimbo tins, exercise books, matchboxes) collected from the school environment.	Two figures are similar if and only if (1) all corresponding angles are equal and (2) all corresponding sides are in the same ratio (the Linear Scale Factor). The scale 1:200 means every length on the plan is 1/200th of the real length, and every measurement on the real building is 200 times the measurement on the plan.	Learners cannot yet fully explain why one tank holds 8 times as much water when it is only twice as tall, or why doubling every length does not simply double the area and volume. These questions are posted on the DQB and will be explained through the Area Scale Factor (Lesson 2) and Volume Scale Factor (Lesson 3) investigations.
Lesson 2 Centre of Enlargement and Linear Scale Factor — Finding the Fixed Point	Learners drew ray lines through corresponding vertices of an object (ABCD) and its enlarged image (A'B'C'D') and observed that all four lines converged at a single point O — the centre of enlargement. They also measured corresponding sides and calculated consistent ratios (image side ÷ object side), including OA'/OA and OB'/OB, all equal to the same L.S.F.	Every enlargement has a unique fixed point called the centre of enlargement (O), from which ray lines through corresponding vertices all pass. The Linear Scale Factor (L.S.F) = image length ÷ object length, and it applies consistently to every pair of corresponding lengths — including distances from the centre O to corresponding vertices. For the Kisumu school plan, L.S.F = 5 cm ÷ 1000 cm = 1/200, which is exactly the '1:200' label on the architectural plan.	The school plan labelled '1:200' was produced using a centre of enlargement and a L.S.F of 1/200: the architect multiplied every real-world length by 1/200 to produce the plan measurement. This is why the 10 m wide classroom block appears as exactly 5 cm on the plan, and why every window, door, and corridor is in exactly the right proportion — the L.S.F is constant for every single measurement in the drawing.
Lesson 3 Constructing Images under Enlargement on a Plane Surface	Learners drew rays from the centre of enlargement O through each vertex of a triangle representing a classroom block on the Kisumu school plan, measured	When a polygon is enlarged with centre O and L.S.F k, every image side equals k times the corresponding object side. A positive k places the image on the same	Learners wrote Claim–Evidence–Reasoning explanations connecting their measurement tables to the formal definition $OA' = k \times OA$, identified the effect of the sign of k on

	distances from O to each vertex, applied the given L.S.F ($k = +2$ and $k = -2$) to locate image vertices, and measured all corresponding sides to verify the ratio.	side of O as the object; a negative k places the image on the opposite side of O and inverts its orientation. The architect's 1 : 200 scale is equivalent to an L.S.F of 1/200 applied to every point of the real building.	image position and orientation, and updated their cumulative Scaling Model diagram to include both positive and negative enlargements anchored to the school architectural plan phenomenon.
Lesson 4 Enlargement on the Cartesian Plane: Positive and Negative Linear Scale Factors	On the Cartesian grid labelled as the Kisumu school coordinate plan, learners observed that when triangle ABC (object in the first quadrant) was enlarged with L.S.F. = 2 from the origin, every image point was exactly twice as far from the origin as the corresponding object point, lying on the same ray. When L.S.F. = -2 was used, the image appeared in the third quadrant — on the OPPOSITE side of the origin — at twice the distance.	Learners learned that: (1) enlargement on the Cartesian plane follows the formula $\text{Image} = \text{Centre} + k \times (\text{Object} - \text{Centre})$; (2) when the centre is the origin, this simplifies to $\text{Image} = k \times \text{Object}$; (3) a positive L.S.F. places the image on the same side of the centre as the object; (4) a negative L.S.F. places the image on the opposite side of the centre, effectively rotating it 180° through the centre, while preserving shape and scaling lengths by $ k $; (5) this is the same mathematical operation used by Kenyan land surveyors assigning GPS grid references to plot boundaries.	The cognitive conflict between prediction and observation — that negative L.S.F. does not merely 'shrink' or 'shift' the image but flips it through the centre — is explained by the vector nature of the enlargement formula. Because L.S.F. scales the VECTOR from the centre to the object point, a negative k reverses the direction of that vector while scaling its magnitude. This is why the image appears on the opposite side at the same scaled distance. This explanation also advances the driving question: lengths scale by k (or $ k $ for negative), confirming and extending the linear scale factor work from Lessons 1–3, and setting up the question of what happens to area (k^2) and volume (k^3) in Lessons 5 and 6.
Lesson 5 Area Scale Factor: Discovering How Area Grows When You Enlarge a Shape	Learners drew pairs of similar rectangles and triangles with L.S.F values of 2, 3, and $\frac{1}{2}$, calculated the area of each figure, and recorded the ratio of the larger area to the smaller area in a structured table.	The ratio of areas of two similar figures equals the square of the linear scale factor: $\text{A.S.F} = (\text{L.S.F})^2$. Doubling all lengths quadruples the area; tripling all lengths multiplies the area by 9; halving all lengths reduces the area to one-quarter.	When the L.S.F between two similar figures is k , every length scales by k , but area — which is calculated by multiplying two lengths together — scales by $k \times k = k^2$. This is why the base of the 2 kg Unga Pembe flour packet has an area 4 times (not 2 times) that of the 1 kg packet, even though each linear dimension is only twice as large. The same principle means the Kisumu school plan at scale 1:200 represents areas at a factor of $200^2 = 40\,000$ smaller than reality.
Lesson 6 Volume Scale Factor and the LSF–ASF–VSF Relationship	Learners calculated volumes of similar solids (sufuria cooking pots, school water tanks, and conical grain stores) and tabulated LSF, ASF, and VSF for three pairs of similar solids with LSF values of 2, 2, and 3.	$\text{VSF} = (\text{LSF})^3$ for any pair of similar solids; the complete three-way relationship is $\text{LSF} = k$, $\text{ASF} = k^2$, $\text{VSF} = k^3$; doubling linear dimensions multiplies volume by 8, not 2 — confirming the school water-tank puzzle from the anchoring phenomenon.	The large school water tank holds exactly 8 times as much as the small tank because $\text{LSF} = 2$ and $\text{VSF} = 2^3 = 8$. The Scale Factor Law (k , k^2 , k^3) governs all similar solids and can be used to calculate unknown capacities in real-life contexts such as water storage tanks, cooking pots, and grain silos across Kenya.
Lesson 7	Learners examined a Survey of Kenya map	Learners confirmed that when a linear scale	Learners used L.S.F, A.S.F, and V.S.F to

Application to Real-Life Situations & Model-Building Project Launch	extract, the school architectural plan (Scale 1 : 200), and a mandazi packaging design problem — each presenting a real Kenyan context where lengths, areas, and volumes are related by L.S.F, A.S.F, and V.S.F respectively.	factor k is applied: area scales by k^2 and volume scales by k^3 — so the taller water tank ($k = 2$) holds $2^3 = 8$ times as much water, not 2 times, resolving the cognitive conflict from the anchoring phenomenon. They applied these relationships to map distances, plan areas, tank volumes, and packaging design.	solve multi-step real-life problems, updated the Driving Question Board to mark the core phenomenon questions as ANSWERED, and produced a fully calculated Project Planning Sheet — including labelled model sketches and a materials list — to guide construction of their own pair of similar solid models in Lesson 8.
Lesson 8 Project Presentations, Model Exhibition & Unit Consolidation — Final Explanation	Groups presented physical models of similar solids (cubes, cuboids, cylinders, spheres) built from locally available materials; learners measured corresponding lengths on each model pair, computed ratios, and displayed their models during a silent gallery walk at desk stations around the classroom.	When a solid is enlarged by a Linear Scale Factor k, the Area Scale Factor is k^2 and the Volume Scale Factor is k^3 ; these relationships explain why the Kisumu school's 1 : 200 architectural plan preserves all proportions while compressing every length by 200, and why a water tank that is only twice as tall holds exactly 8 times as much water as the smaller tank.	Learners constructed individual cumulative annotated diagrams labelling L.S.F, A.S.F, and V.S.F on a cylinder object-image pair, wrote a driving-question paragraph connecting the three scale factor rules to the school plan and water tanks, completed Final Answer Cards with a novel Kenyan real-life example, and recorded a reflective 'I used to think / Now I know' entry documenting their conceptual journey from Lesson 1 intuition to Lesson 8 proof.

ATTRIBUTION AND LICENSING

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